

Set-A

 BRAC University Department of Computer Science and Engineering Semester: Spring 2026 Course Code: CSE460 Course Title: VLSI Design	Midterm Examination Full Marks: 10 x 2 + 15 = 35 Time: 1 hour 20 minutes Date: 30 March 2026
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Student ID:	Name:	Section:
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[Answer all **THREE** questions. **Return** the question paper with the answer script.]

Question 1: [CO2]

Simon is required to design a sequence detector that can detect the patterns **101** and **110** from a serial data stream. When the sequence 101 is detected and the Red LED (R) turns on; when 110 is detected, the Blue LED (B) turns on.

You may use the variables **R** and **B** to represent the status of the corresponding LEDs.

A sample test is given below to show the detector circuit's functionality:

Clk	1	2	3	4	5	6	7	8	9	10	11	12
w	1	0	1	1	0	0	1	1	0	1	0	0
R	0	0	1	0	0	0	0	0	0	1	0	0
B	0	0	0	0	1	0	0	0	1	0	0	0

(a)	Determine the type of FSM that would be needed for the given problem and design the state diagram for the corresponding FSM. Using more states than necessary will result in a deduction of marks.	[5]
(b)	Derive the state-assigned table using the Binary encoding from your state diagram in (a).	[3]
(c)	Find the expression for the outputs (R and B) for the FSM using KMAP.	[2]

Question 2: [CO2]

A digital communication system transmits data bits serially through a channel. To enable simple error detection at the receiver, a **parity bit** is generated for every **group of three** transmitted bits. A circuit receives a serial input w, which represents the incoming data bits. After observing **three consecutive bits**, the circuit generates an **odd parity bit P**. The parity bit is defined as:

- P=1 if the number of **1s** in the three-bit sequence is **odd**
- P=0 otherwise

After producing the parity bit for three consecutive incoming bits, the circuit immediately starts processing the **next group of three incoming bits**.

The table below shows a sample operation:

w	0	1	0	0	1	1	1	0	0	1	1	1	1	0	1	1
P	0	0	0	1	0	0	0	0	0	1	0	0	1	0	0	0

(a)	Design the state diagram for the corresponding FSM with a minimum number of states . How many Flip-Flips are required to implement the circuit? [Hint: To minimize the number of states, different input sequences that lead to the same combination (number of bits observed and parity so far) can share the same state.]	[6]
(b)	Derive the state-assigned table from the state diagram using Binary Encoding .	[3]
(c)	Find the expression for the output logic for the FSM using KMAP.	[1]

Question 3: [CO1]

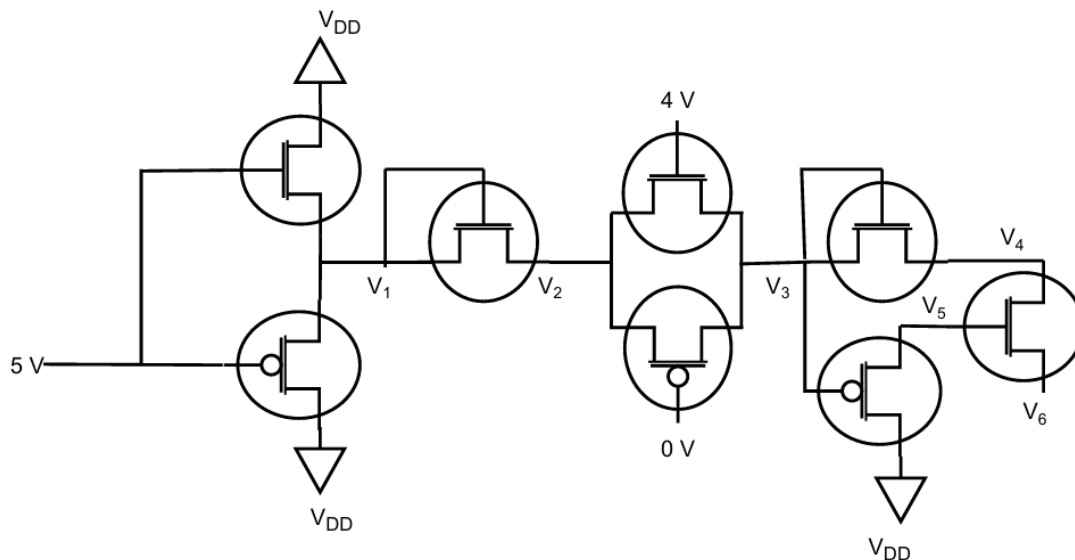


Figure 1

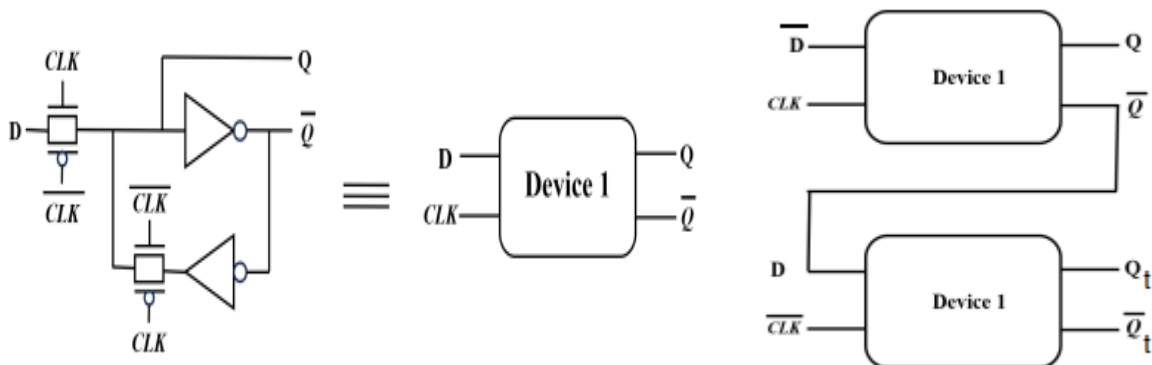
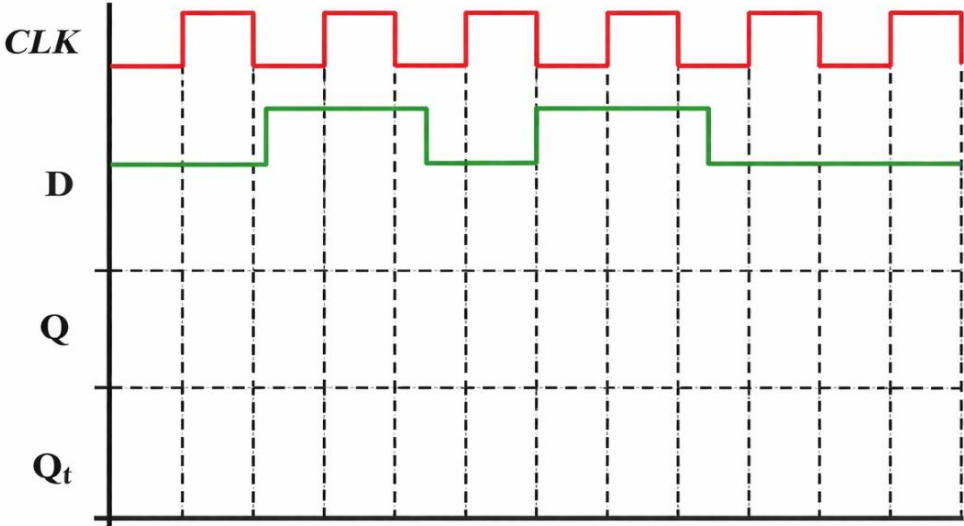


Figure 2

(a)	Suppose you are tasked with designing a 4:1 mux. The client requires the design of the mux to be such that even if the input is a weak signal, the output is strong. Design the mux and specify the number of transistors used. [Notes: All the input signals are also available in its complement form]	[5]
(b)	If $V_{DD} = 5V$, $V_{tn} = 0.5V$ and $V_{tp} = 0.6V$, find the node voltages from Figure 1.	[5]
(c)	For the circuit in Figure 2, draw the waveform of Q and Q_t and identify the circuit type. 	[5]